which on exposure to the atmosphere absorb moisture from atmosphere, dissolve in the same and change into a solution.
Efflorescence - Crystalline hydrated salts which on exposure to the atmosphere lose their water of crystallisation partly or completely and change into a powder.

(iv)
(a) Tribasic 7
(b) Dibasic 12
(c) ~~3 to 7~~ 13
(d) ~~5~~ 3 to 7
(e) 5
(f) 10

one atom should have a lone pair while one atom should lack a lone pair. Example- Hydronium (H_3O).

4 (i)
(a) ~~[illegible]~~ ~~2~~ 2,8,8,2
(b) ~~Ionic compound~~ group 2, period 4

(ii) (a) 2
(b) ionic

(iii) Avogadro's law states that under the same conditions of temperature and pressure equal volumes of all gases contain the same number of molecules. Three applications are :-
→ Determines the atomicity of the gas
→ Determines the molecular formula of a gas
→ Explains gay-lussac's law.

(iv) ~~B~~
(i) A, B
(ii) A, C
(iii) D, B

5
(i)
(a) Gypsum
(b) Baking soda
(ii) Deliquescence - water soluble salts

3

(I) When it reacts with Sulphur it will form MS.
When it reacts with Chlorine it will form MCl_2.

(II) :

(a) Hydrogen atoms form only one covalent bond because they only have one valence electron to pair. On the other hand, oxygen needs two ~~atoms~~ electrons to complete its octet. Since each atom in an oxygen molecule shares two electrons with its partner, we get a covalent ~~bond~~ double bond.

(b) Carbon tetrachloride molecules have 4 single bonds as the carbon atom is attached to each chlorine atom by different single covalent bonds.

(III)

(i) $MgCl_2$ – $Mg: + 2\,\ddot{Cl}: \rightarrow [Mg]^{2+} + 2[Cl]^{1-} \rightarrow MgCl_2$

(ii) CaO – $Ca: + \ddot{O}: \longrightarrow [Ca]^{2+} + [O]^{2-} \rightarrow CaO$

(iii) NaCl – $Na\cdot + \ddot{Cl}: \longrightarrow [Na]^{1+} + [Cl]^{1-} \longrightarrow NaCl$

(IV) It is a bond formed by a shared pair of electrons with both electrons coming from the same atom. The conditions are that

Section B

2

(i) 1 False. The cations take the electronic configuration of the noble gas closest to it and is always the noble gas in the period above.

2 True. This is because an ionic compound is formed which is a good conductor of electricity.

(ii) Molar mass of 1 mole of CO_2 = 12+2(16) = 44

44 gm of CO_2 = 1 mole of CO_2

So,

22 gm of CO_2 = $\frac{22}{44}$ moles of CO_2 = 0.5 moles of CO_2

(iii)

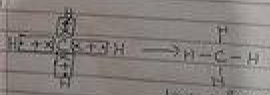
(a) Oxidising power of X will be greater

(b) X will have more electronegativity

(c) X will be towards the right of Y.

(iv)

(i)



four single covalent bonds

(ii)

(a) Barium as it has least ionisation potential

(b) They all have 2 valence electrons.

(iii) $C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$

RMM of propane = 44

44 g of propane requires 5×22.4 l of oxygen

8.8 g of propane requires $\frac{5 \times 22.4 \times 8.8}{44}$

=> 22.4 litres

(iv)

(i) number of gram atoms = $\frac{\text{weight}}{\text{molecular weight}}$

=> $\frac{4.6}{23}$

=> 0.2 g atoms

(ii) molecular weight of $H_2O = 2(1) + 16 = 18$ g/mol

molecular weight of $CuSO_4 \cdot 5H_2O = 64 + 32 + 4(16) + 5(18)$

$= 250$ g/mol

% of water of crystallization in $CuSO_4 \cdot 5H_2O$ =

$\frac{5 \times 18}{250} \times 100 = 36\%$

(iii) VD = 2EFW

MW = 2VD

MW = 2EFW

$2 \times XY_2 = X_2Y_4$

(v)

(i) 15

(ii) 19

(iii) 8

(iv) 4

(v) 2

16.

(i)

(a) Z

(b) L

(c) Halogens

(d) Nitrogen, Boron

(e) O

(ii) (a) Reducing agents

(b) lead

(c) acidic

(d) Silver chloride

(e) Cl

(iii)

(i) Atomic mass : S = 32 and O = 16

molecular mass of $SO_2 = 32 + (2 \times 16) = 64g$

As 64g of SO_2 = $22.4\ dm^3$

Then,

$320g$ of $SO_2 = \frac{320 \times 22.4}{64} = 112\ L$

(ii) Gay Lussac's law states that when gases react, they do so in volumes which bears a simple whole number ratio to one another and to the volumes of the products, if gaseous, provided the temperature and pressure of the reacting gases and their products remain constant.

Chemistry
Section A

1. (4) Argon
2. (1) Potassium
3. (4) 22
4. (1) 17
5. (3) 1:1
6. (4) Mercury
7. (2) lead
8. (1) 1
9. (1) Lithium
10. (4) Ammonium chloride
11. (4) Copper chloride
12. (4) Conducts electricity when it is in molten state
13. (3) An Alkali
14. (1) Direct combination
15. (3) conc. HCl + conc. HNO_3